



Paper and code

Short-to-Long Functional Connectivity Transfer via Structure-Aware Latent Diffusion

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The University
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at Chapel Hill

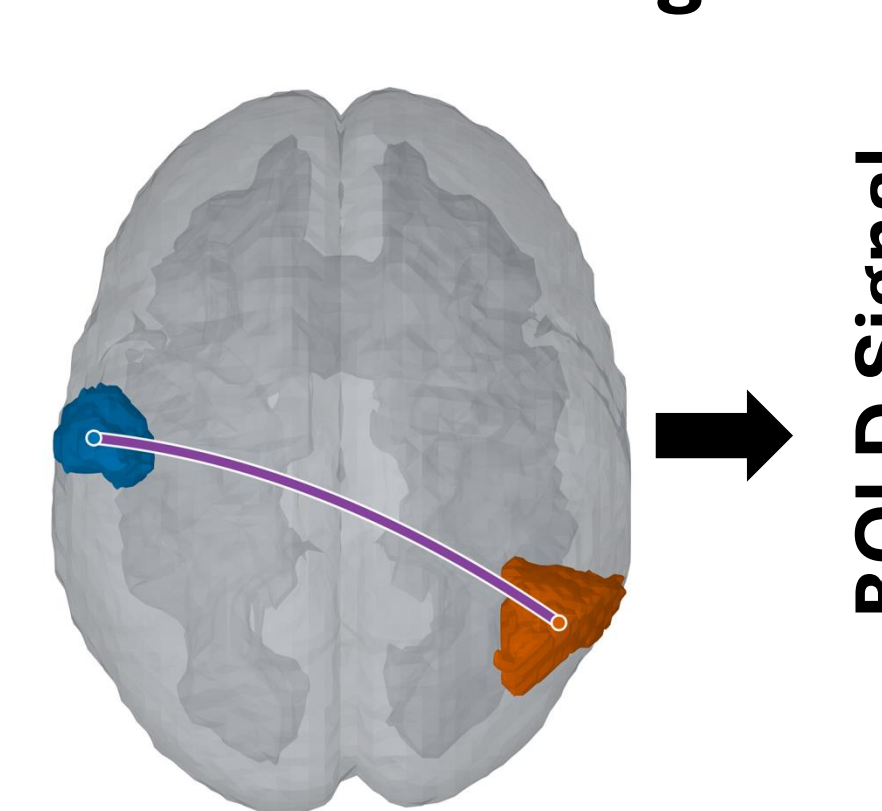


Motivation, Task, and Challenge

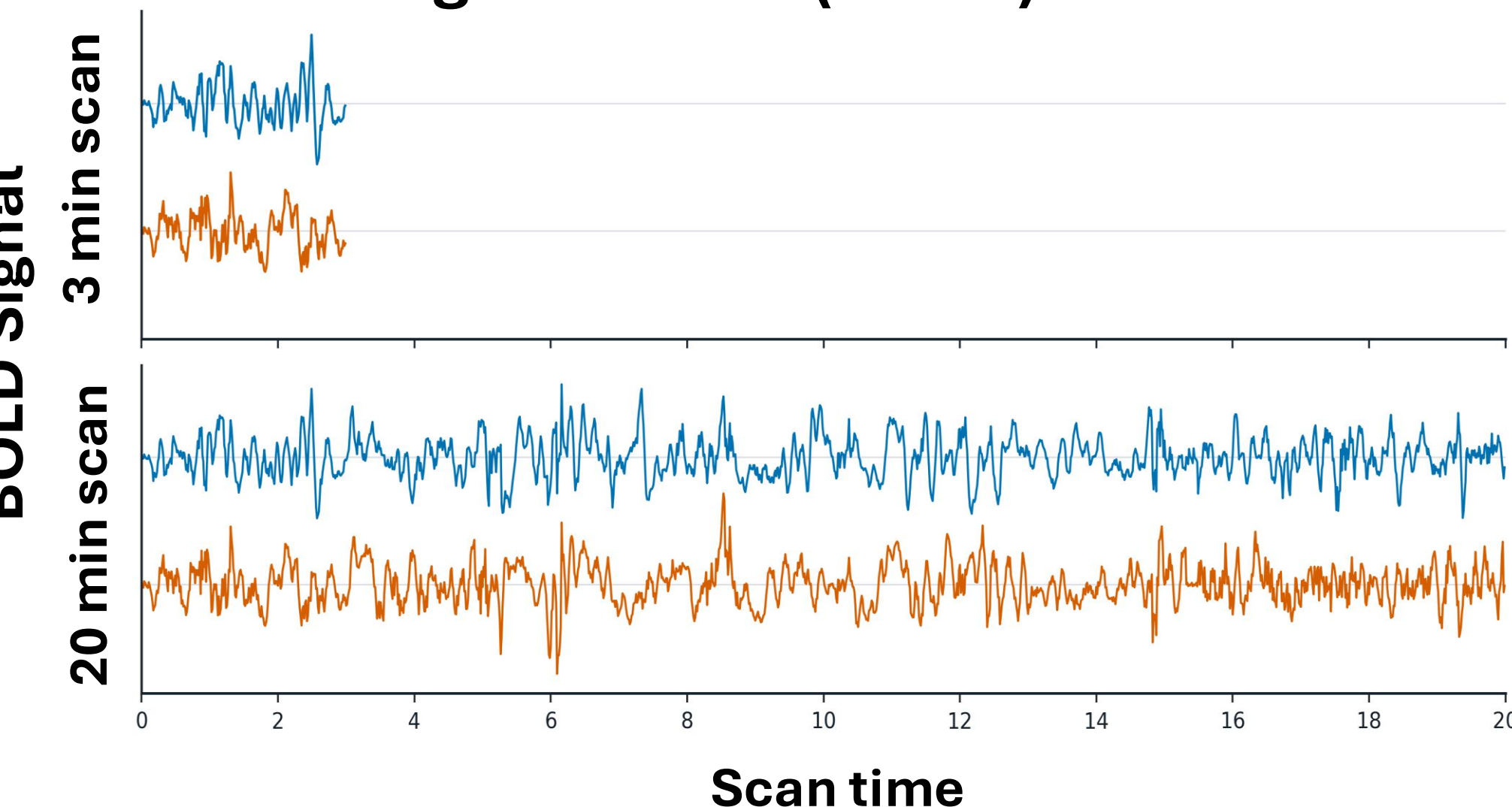
Motivation

Short fMRI scans give noisy estimates of how brain regions interact. Longer scans improve these estimates and behavioral prediction but cost more and burden participants.

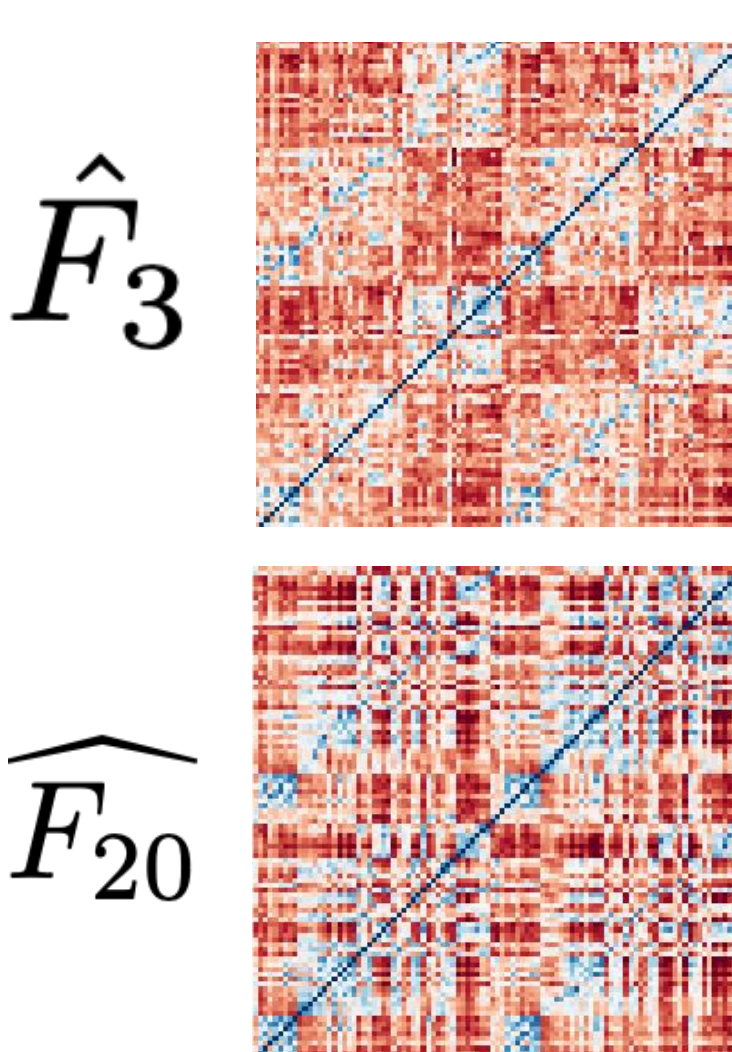
Coordinated activity
across brain regions



Regional fMRI (BOLD) Time Series



Duration-Dependent
Connectivity



Task

Generate long-scan brain connectivity from a short scan, anatomy, and subject information.

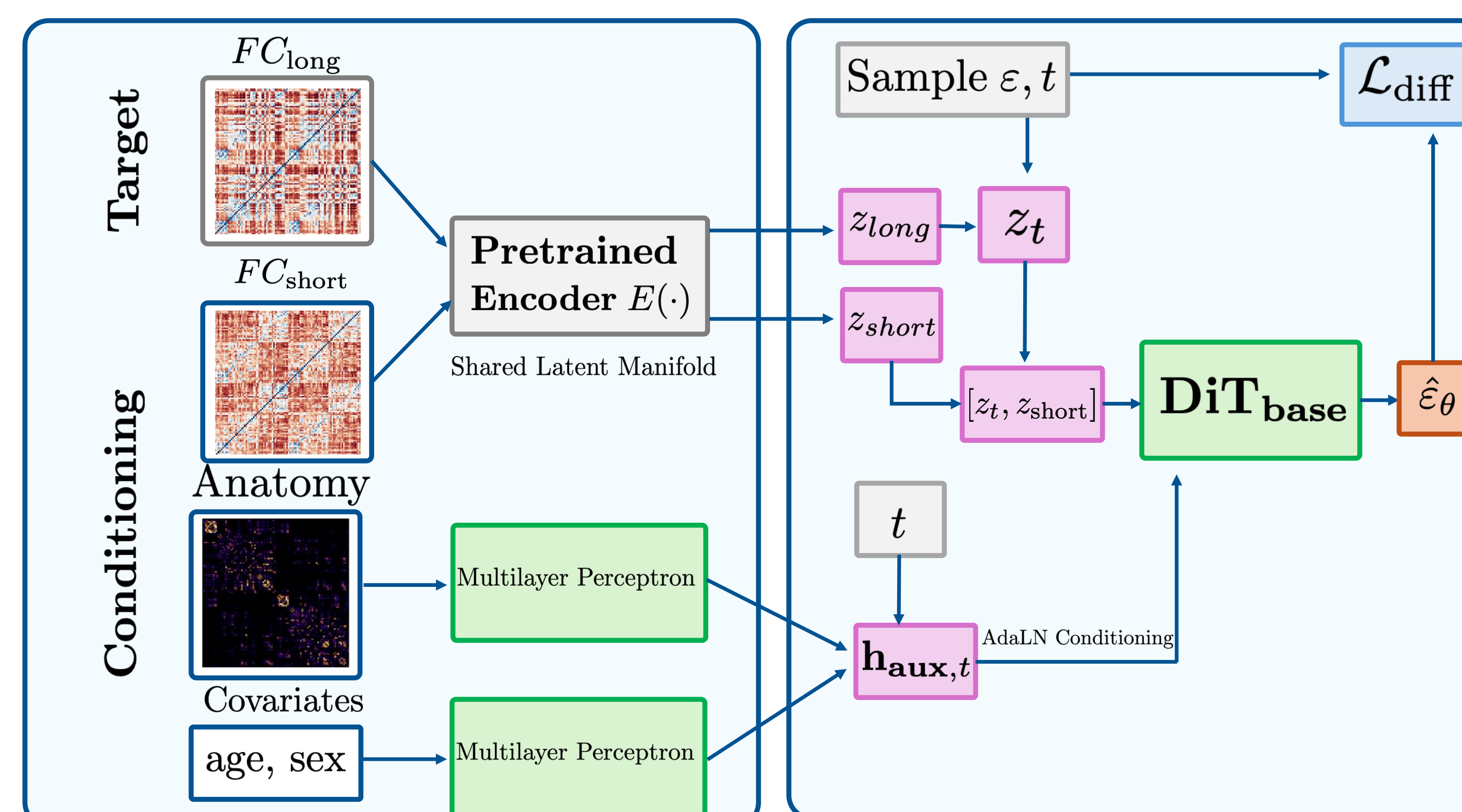
Challenge

Improve reconstruction without erasing individual differences related to cognition and behavior.

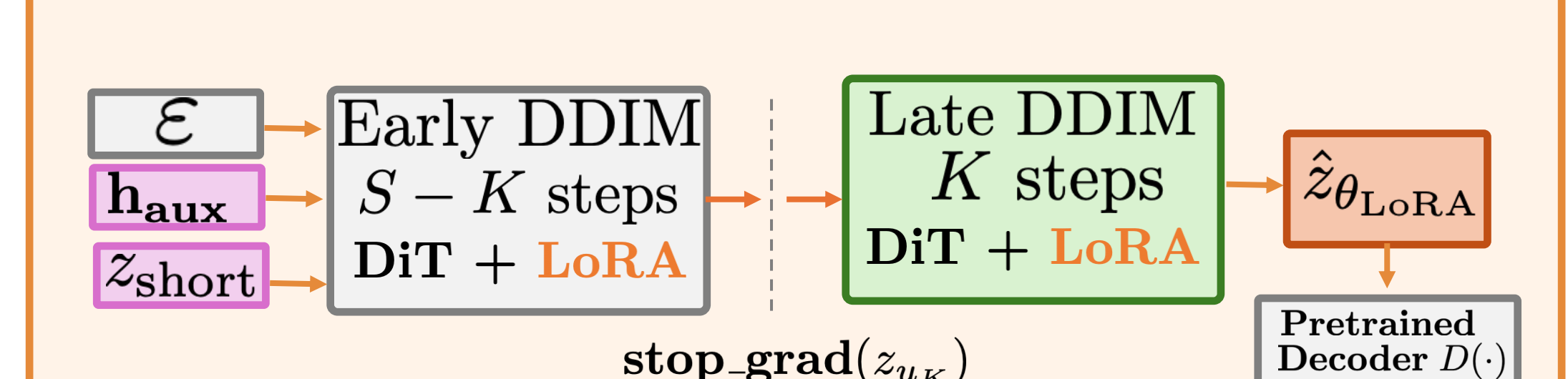
Structure-Aware Latent Diffusion with Phenotype Steering

SALD builds on direct reward fine-tuning (DRaFT): phenotype loss is backpropagated through the final (K) DDIM steps to update only LoRA.

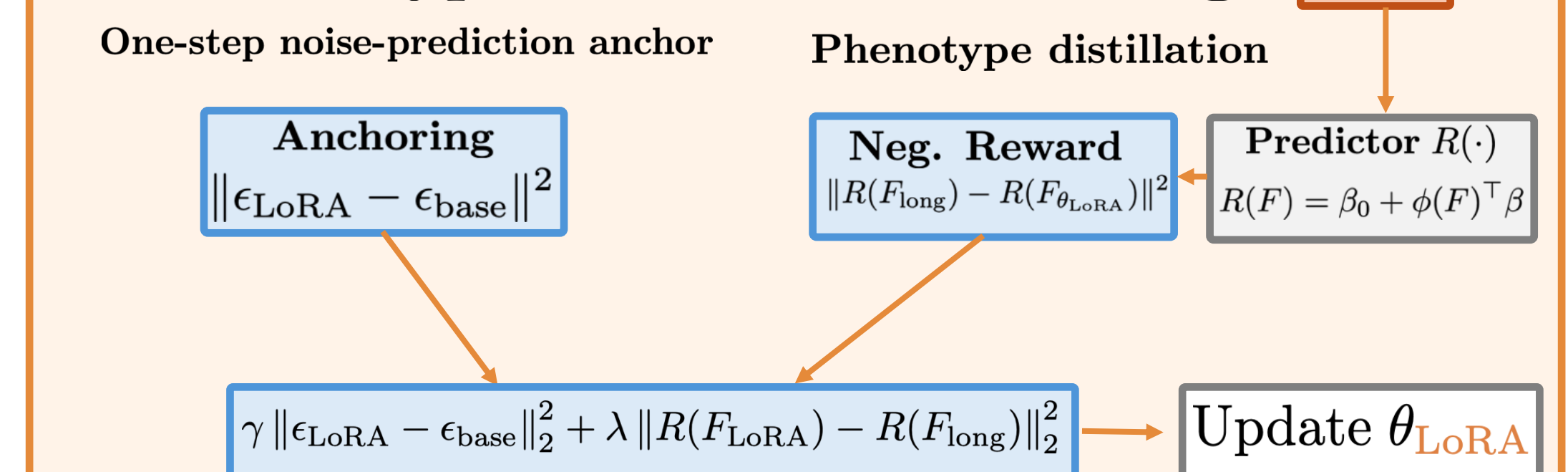
- Inputs and latent encoding**
- Base diffusion training**
One random timestep
one noise prediction
- SALD Sampling Steering**



A. DDIM Sampling



B. Phenotype-Aware Loss Steering

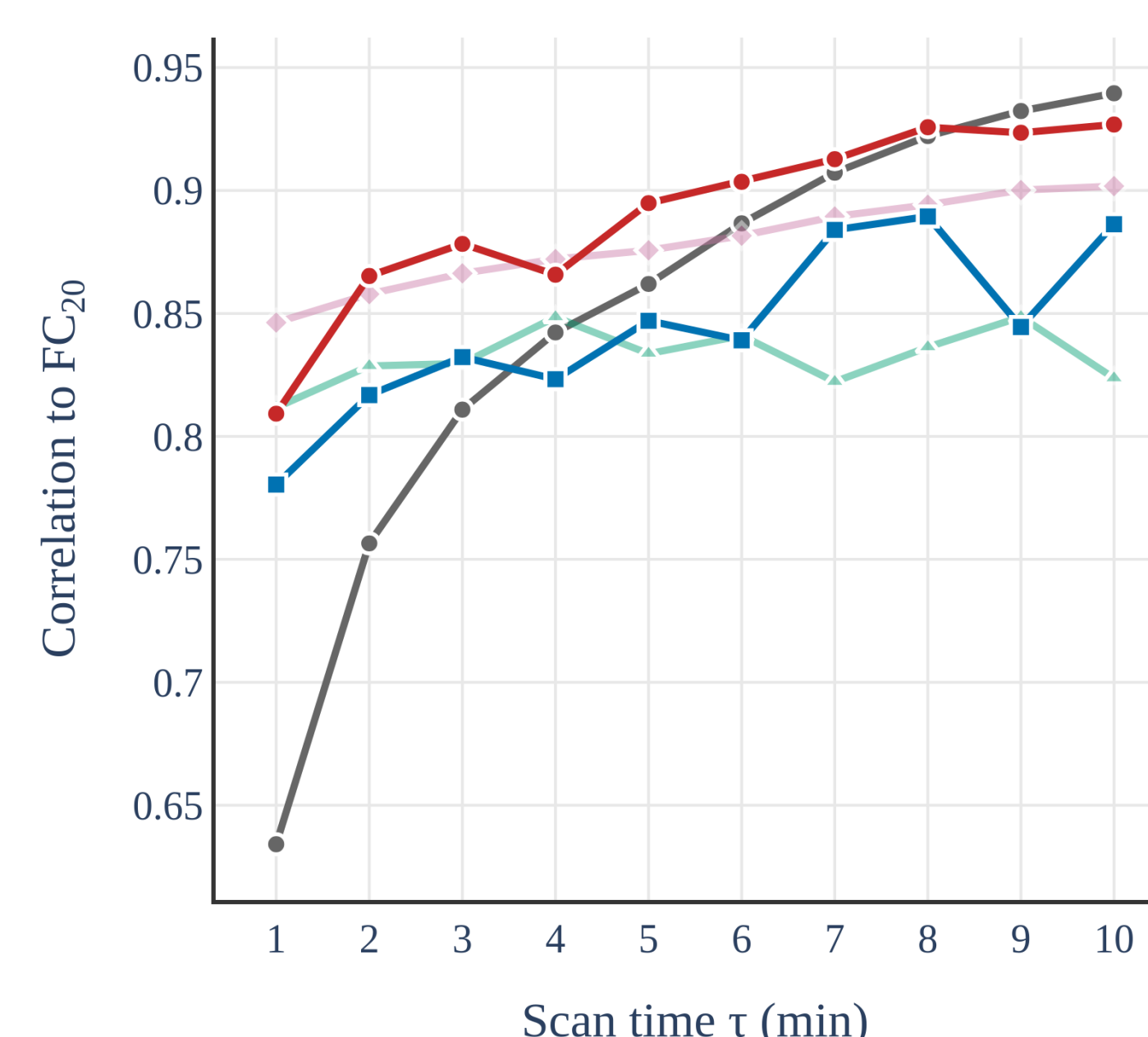


Results

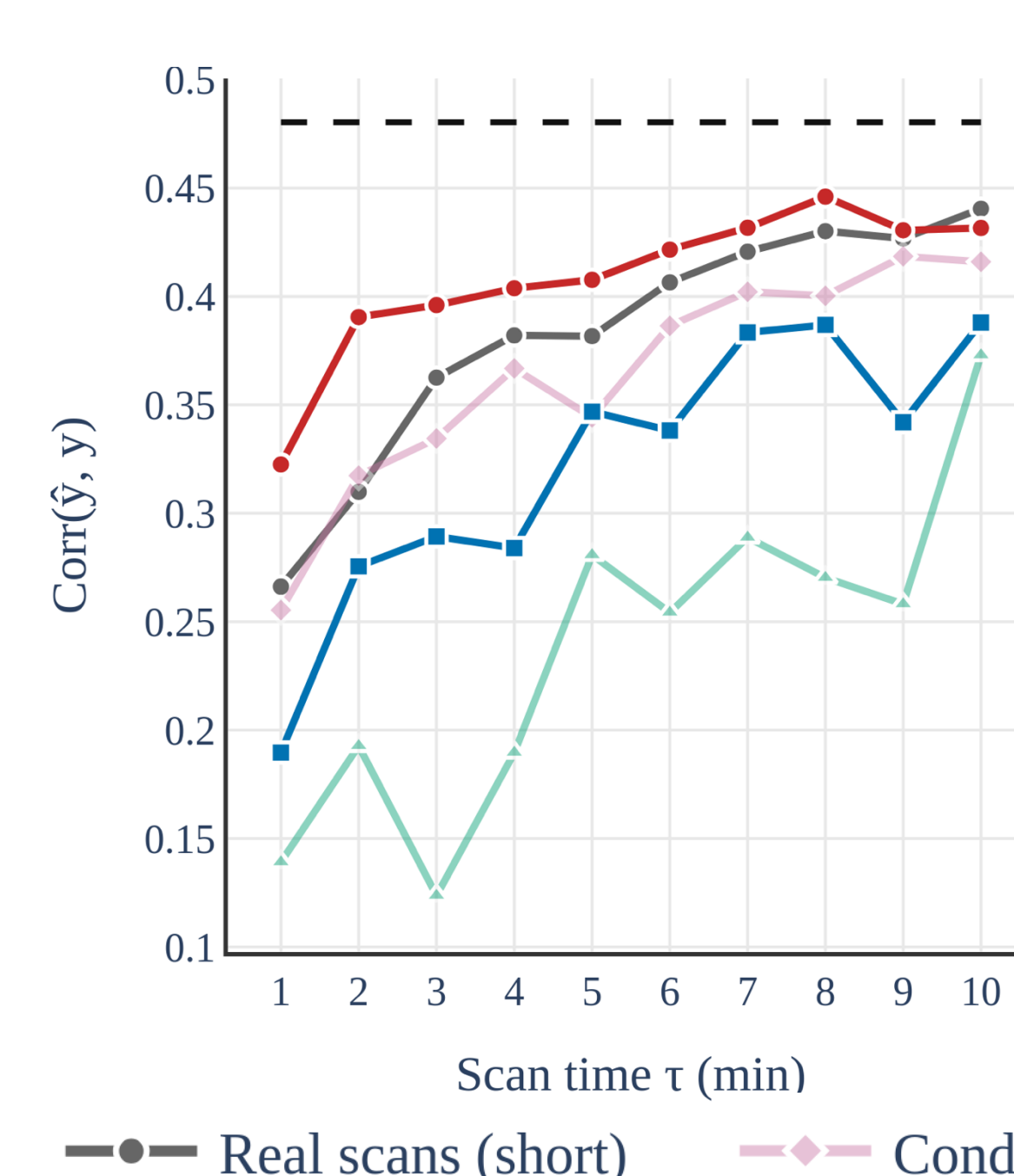
Steering improves phenotype recovery and differential identifiability

Across 11,156 ABCD scans from children ages 9 to 12, SALD improves recovery of phenotype-relevant variation and differential identifiability relative to diffusion without steering. Gains are largest at short scan times and extend to phenotypes not used during steering, including impulsivity, psychopathology, and sleep disturbance.

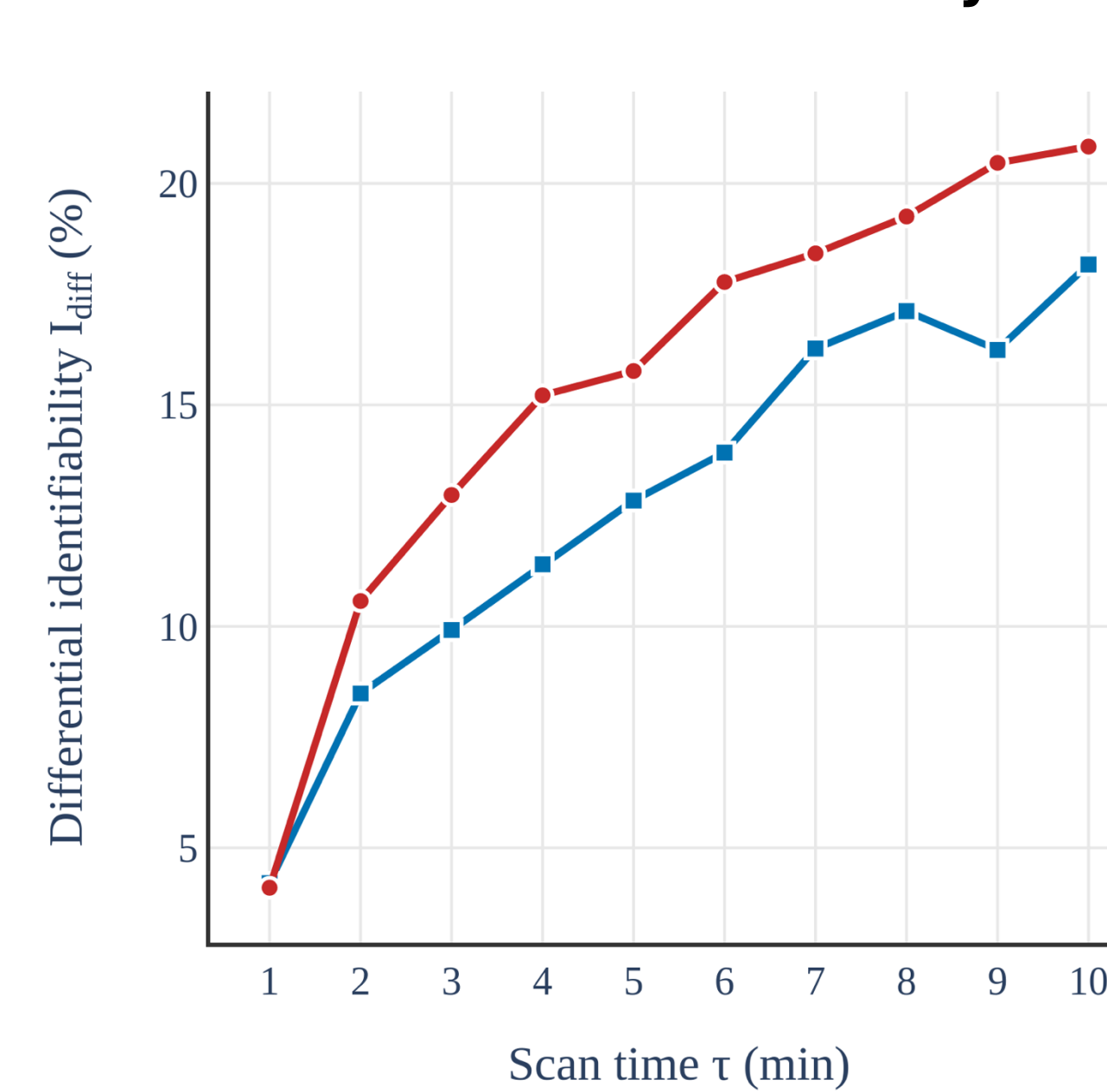
Does it look right?
Similarity to the 20-min
reference



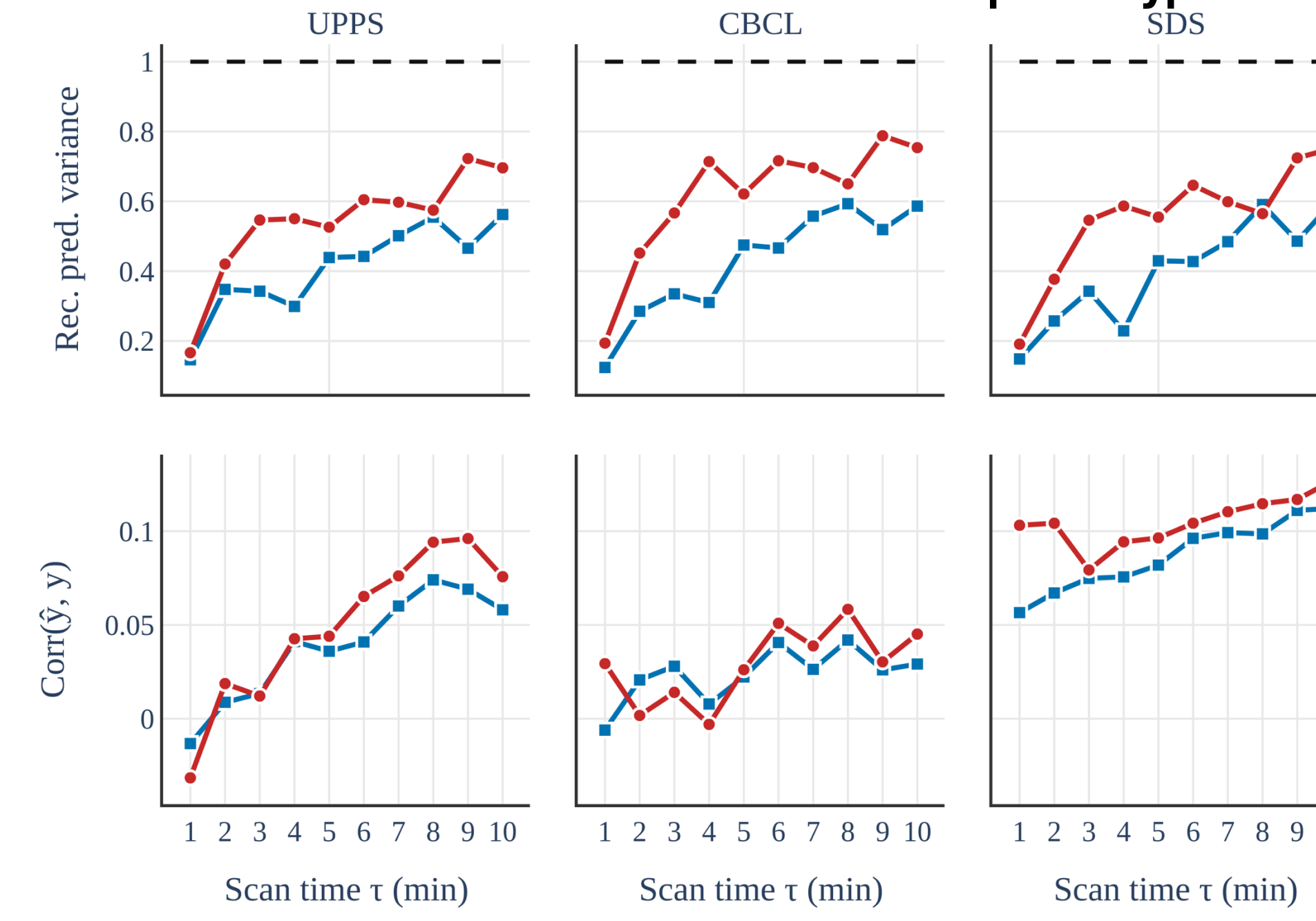
Is it still useful?
Cognition prediction



Is it still the same
subject?
Differential identifiability



Does it generalize?
Prediction on unseen phenotypes



Conclusions and Limitations

Conclusions

- SALD synthesizes **long-scan-like FC from short acquisitions**.
- Likelihood-based diffusion can **attenuate phenotype-relevant variation**.
- Reward alignment **improves recovery of this variation**, with the **largest gains at short scan times**.

Limitations

- Evaluated on a **single ABCD configuration**; external validation is needed.
- Broader reliability**, including test-retest behavior, remains to be evaluated.
- SALD targets **population-level research, not clinical decision-making**.